# Bioinformatics session

4th annual workshop on bioinformatics and variant interpretation in InPreD

https://inpred.github.io/26-09_bioinfo_ws/

# 2. Containerization

## A brief history

| | |
|---|---|
| **1979** | `chroot` system call (changing the root directory of a process and its children to a new location in the filesystem) in Uni V7 considered as beginning of process isolation |
| **2000** | FreeBSD Jails allows administrators to partition FreeBSD computer system into several independent, smaller systems (*jails*) – with the ability to assign IP address for each system and configuration |
| **2001** | Linux VServer is jail mechanism that can partition resources, similar to FreeBSD Jails |

| | |
|---|---|
| **2004** | First public beta of Solaris Containers released - combines system resource controls and boundary separation provided by zones |
| **2005** | Open Virtuzzo is operating system-level virtualization technology for Linux which uses patched Linux kernel for virtualization, isolation, resource management and checkpointing |
| **2006** | Process Containers, launched by Google, was designed for limiting, accounting and isolating resource usage of collection of processes - renamed to *Control Groups (cgroups)* and merged into Linux kernel |
| **2008** | LinuX Containers (LXC) was first, most complete implementation of Linux container manager - implemented using cgroups and Linux namespaces |

## A brief history

| | |
|---|---|
| **2011** | CloudFoundry started Warden can isolate environments on any operating system running as daemon and providing API for container management |
| **2013** | Let Me Contain That For You (LMCTFY) kicked off as open-source version of Google’s container stack - providing Linux application containers |
| **2013** | Docker emerged |
| **2016** | Singularity (later Apptainer) was released |

## Docker

### What is it? 🔎

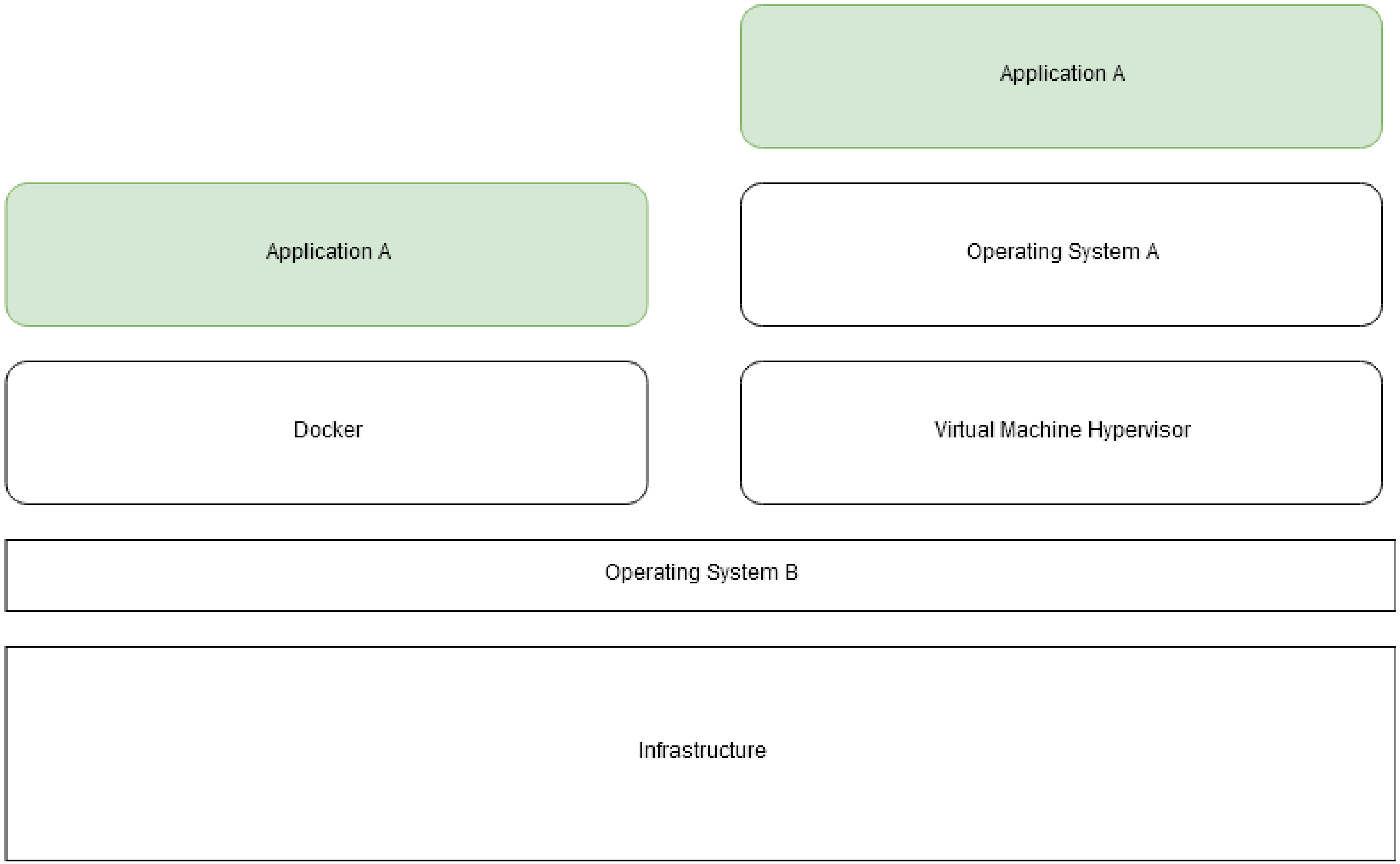
- containerization technology to package application and dependencies into a container
- the container image can be shipped and run consistently across different computing environments

**Which benefits does it provide?**
🎁

1. Consistency
2. Isolated environments
3. Portability
4. Efficiency

## How is it different from virtual machines? 💻

- containers use and share host OS's kernel making them more lightweight and efficient
- virtual machines emulate entire physical machine, including OS making it possible to run multiple OS instances on single physical machine


Application A
Application A
Operating System A
Docker
Virtual Machine Hypervisor
Operating System B
Infrastructure

## Key Concepts 🔑

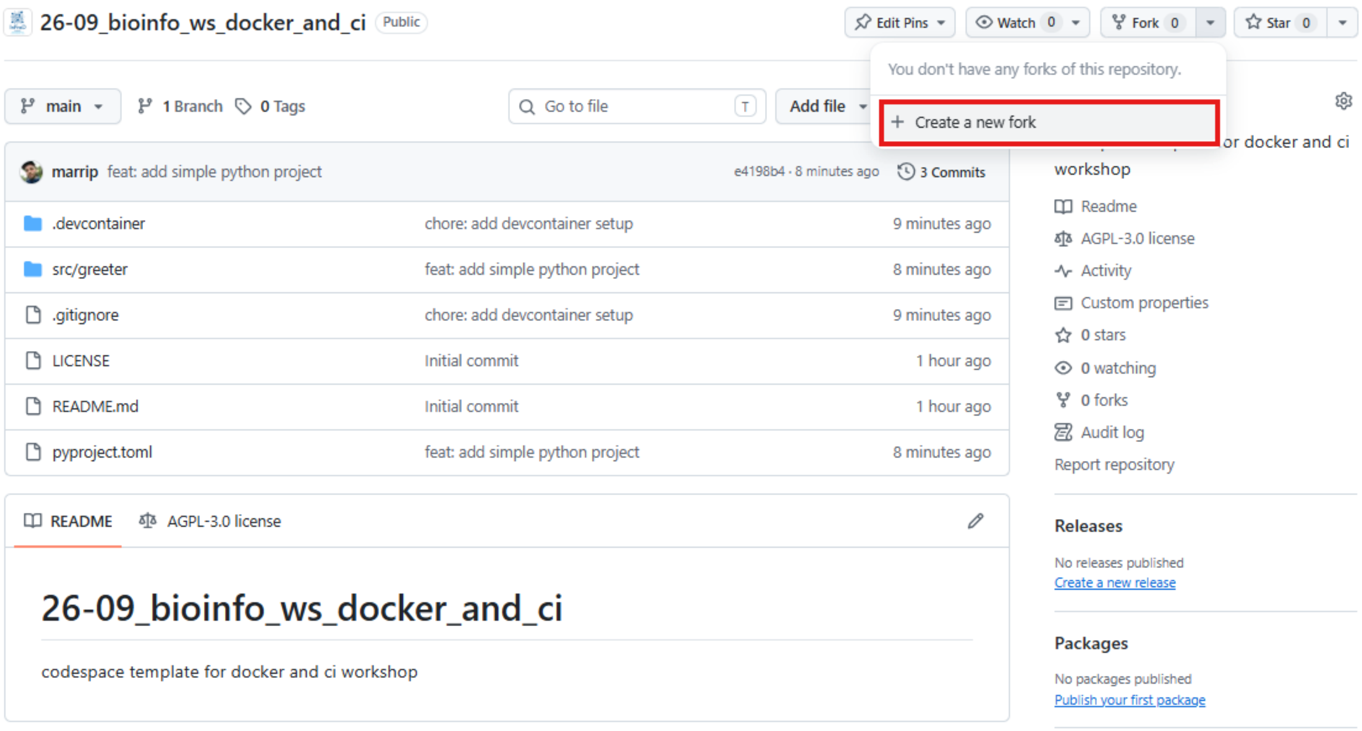
1. **Image:** standardized package that includes all files, binaries, libraries, and configurations to run container
2. **Container:** running instance of image providing isolated runtime environment
3. **Dockerfile:** instructions on how to build image
4. **Docker Hub:** public registry where developers can share and access pre-built images

## Key Components 🔑

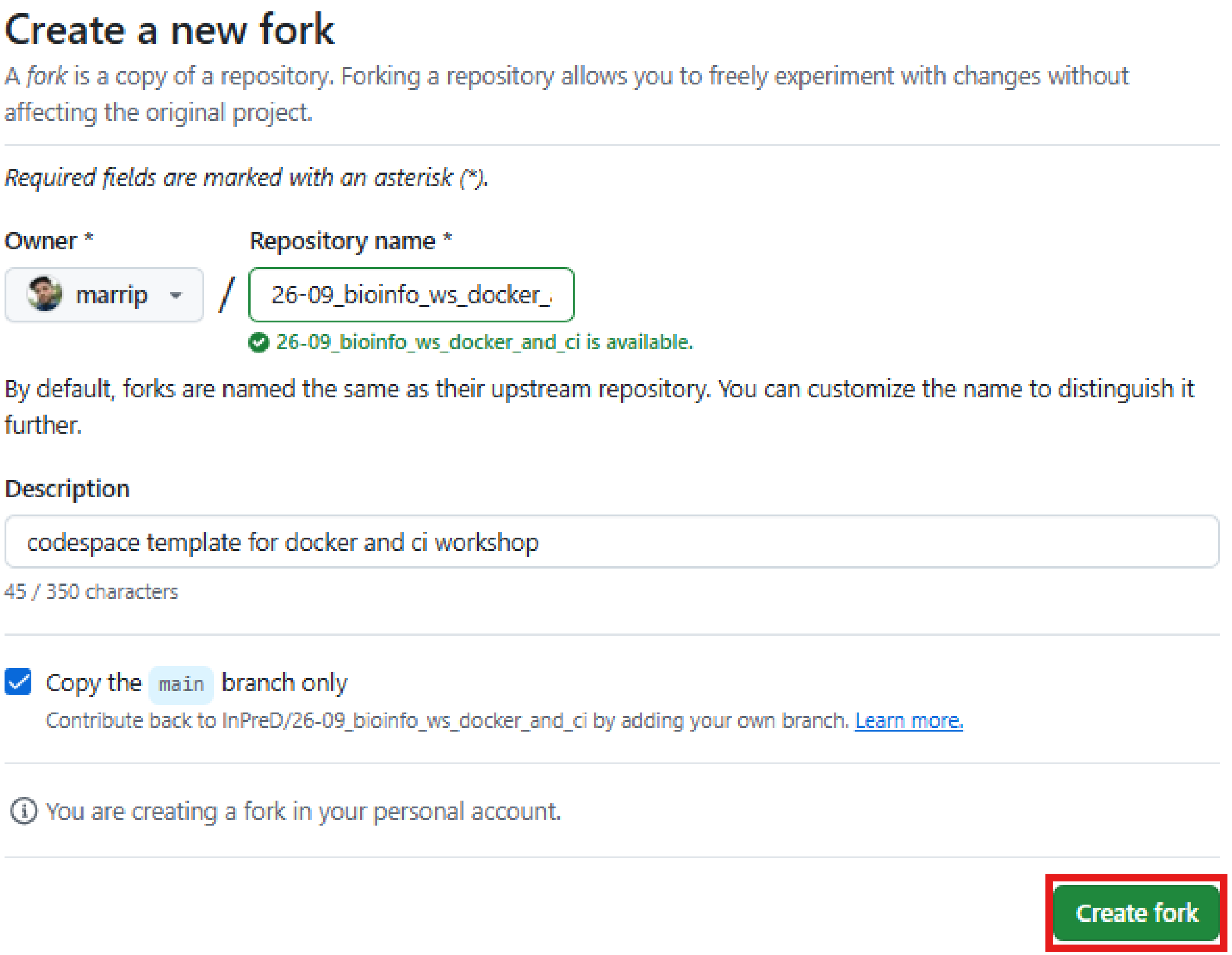
1. **Docker Engine:** core, open-source technology that builds and runs containers
2. **Docker Client:** command-line tool that sends instructions to the Docker Daemon using REST APIs
3. **Docker Daemon (`dockerd`):** receives and processes API requests and calls container runtime
4. **Container Runtime (`containerd`):** turns static container image into running, isolated application on host OS; industry standard

# Let's explore! 🗺️

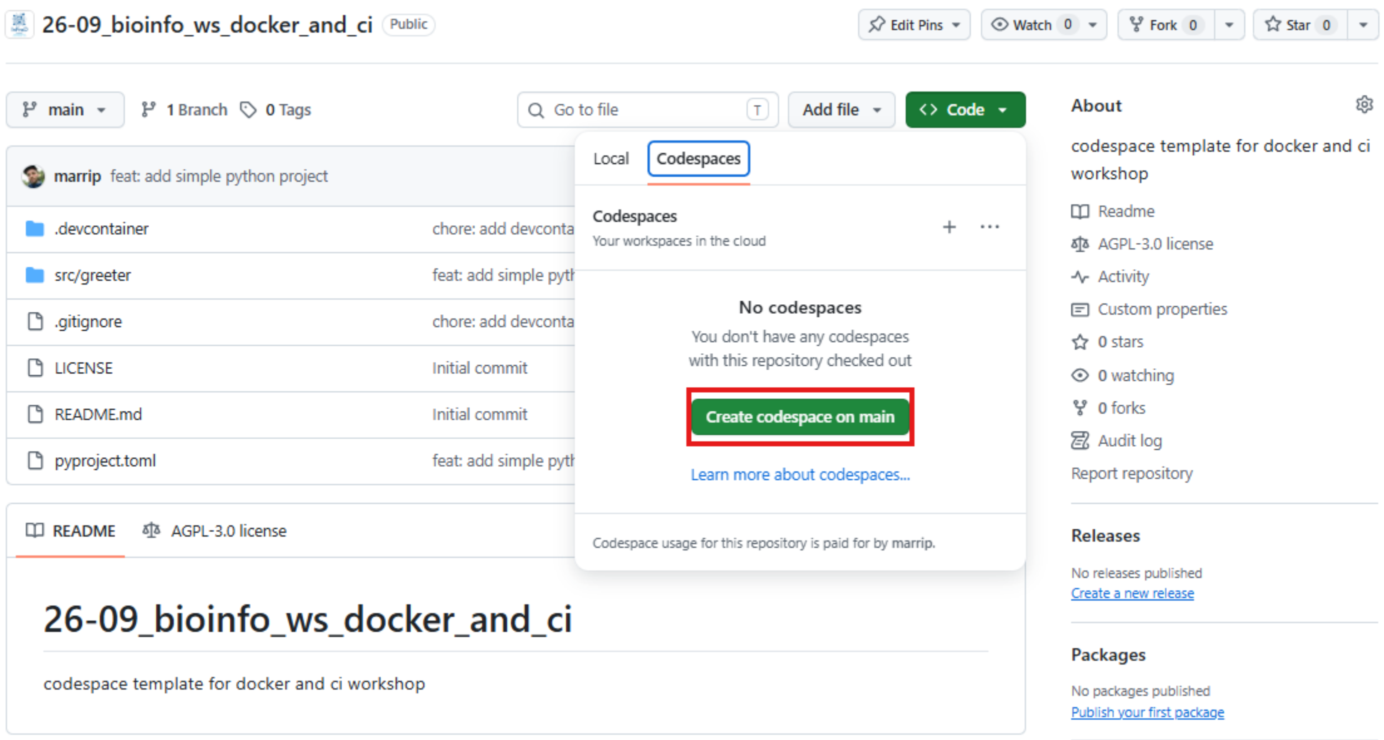
Start by going to https://github.com/InPreD/26-09_bioinfo_ws_docker_and_ci and create a fork



Create your own fork by clicking on `Create fork`



In the forked repository, navigate to `Code` > `Codespaces` > `Create codespace on main`



# 3. Continuous Integration (CI)